ReactJS

Unit 4 — Conditional Rendering · Lists · Rendering lists · Keys

In This Unit

1. Conditional Rendering
2. Rendering lists with map()
3. Keys and why they matter

CONDITIONAL RENDERING

if/else · Ternary · Show/hide

What is Conditional Rendering?

- Conditional rendering = showing or hiding content based on a condition
- Like an if/else statement in JavaScript
- Common in React: logged in? show Dashboard, else show Login

```
if (isLoggedIn) {
  show Dashboard
} else {
  show Login
}
```

Method 1: Ternary Operator (?:)

```
const [loggedIn, setLoggedIn] = useState(false);

<div>
  {loggedIn ? (
    <h1>Welcome back!</h1>
  ) : (
    <h1>Please log in</h1>
  )}
</div>
```

- condition ? (if true) : (if false)
- Most common pattern in React
- Can inline in JSX

Method 2: Logical AND (&&)

```
const [showMessage, setShowMessage] =
  useState(true);

<div>
  {showMessage && (
    <p>This only shows if true</p>
  )}
</div>
```

- True && show content
- False && show nothing
- Good for simple show/hide

Method 3: if/else Before Return

```
function Login() {
  const [loggedIn, setLoggedIn] =
    useState(false);

  if (loggedIn) {
    return <h1>Welcome!</h1>;
  }

  return <h1>Log in first</h1>;
}
```

- Use if/else before the return statement
- Also called early return
- Good for complex logic

Example: Toggle Button

```
function Toggle() {
  const [show, setShow] = useState(false);

  return (
    <div>
      <button onClick={() => setShow(!show)}>
        {show ? 'Hide' : 'Show'}
      </button>
      {show && <p>You can see me!</p>}
    </div>
  );
}
```

Conditional CSS Classes

```
const [active, setActive] = useState(false);

<div className={active ? 'active' : 'inactive'}>
  Content
</div>

{/* or using && */}
<div className={'btn ' + (active && 'btn-active')}>
  Content
</div>
```

Practice

- Create a Counter with +/- buttons
- Make a TextInput that shows what you type
- Build a Form with name and email fields
- Use preventDefault() to stop page reload
- Track state with useState

Summary

- State = component data that changes
- useState = Hook to track state in function components
- Events are camelCase: onClick, onChange, onSubmit
- Controlled components use React state for input values
- Call e.preventDefault() in form submit handlers

LISTS

map() · Rendering arrays

Rendering Lists with map()

- To render a list from an array, use map()
- map() transforms each item into JSX
- The browser displays an HTML list

```
const cars = ['Ford', 'BMW', 'Audi'];

<ul>
  {cars.map((car) => (
    <li>{car}</li>
  ))}
</ul>
```

Rendering Components in a List

```
function Item(props) {
  return <li>{props.name}</li>;
}

function List() {
  const items = ['Apple', 'Banana', 'Cherry'];

  return (
    <ul>
      {items.map((item) => (
        <Item name={item} />
      ))}
    </ul>
  );
}
```

List of Objects

```
const students = [
  {id: 1, name: 'Ravi'},
  {id: 2, name: 'Kiran'},
  {id: 3, name: 'Anil'}
];

{students.map((s) => (
  <li>{s.name}</li>
))}
```

- Use objects to store data
- Access properties with dot notation

KEYS

Helping React track items

What are Keys?

- Keys help React track items in a list
- Without keys, React doesn't know which item changed
- If you delete one item, React re-renders the entire list
- With keys, React re-renders only that item

Using Keys: Example

```
const students = [
  {id: 1, name: 'Ravi'},
  {id: 2, name: 'Kiran'},
  {id: 3, name: 'Anil'}
];

{students.map((s) => (
  <li key={s.id}>{s.name}</li>
))}
```

- Pass the key prop to the outer element
- Use a unique id for best results
- Keys help React optimize performance

Key Rules

Rule	Explanation
Unique among siblings	No two items share the same key
Stable	Same item = same key every render
On outer element	Put key on what map() returns
Not a prop	key is not passed to children

When Index as Key is OK

- Use array index as key only when:
- The list is static (never changes)
- The list is never filtered or reordered
- Items have no real IDs

Not recommended for dynamic lists

```
{items.map((item, index) => (
  <li key={index}>{item}</li>
))}
```